

L-functions for all meromorphic $\mathrm{SL}_2(\mathbb{Z})$ modular forms, via manifestly finite contours

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Abstract

We give a single manifestly finite, regulator-free functional realizing in one object L-functions quasi-periods and period polynomials of meromorphic modular forms on $\mathrm{SL}_2(\mathbb{Z})$, with generic poles throughout the upper half-plane $\mathbb{H}$ and at the cusp of $\mathrm{SL}_2(\mathbb{Z})$. Additionally, our construction preserves holomorphicity. It is a two-segment contour integral, $L^*(f, s) = L_T^*(f, s; \gamma_{\tau_0}^T) + L_S^*(f, s; \gamma_{\tau_0}^S)$, where L_T^* and L_S^* have distinct integration kernels, and $\gamma_{\tau_0}^S$ is a path in $\mathbb{H}$ from $S(\tau_0)$ to τ_0 , and the same for $\gamma_{\tau_0}^T$. For weakly holomorphic forms our L-function is independent of τ_0 and of the path, and reproduces classical L-functions for holomorphic modular forms, reproduces the L-functions of weak modular forms in [1], which previously needed non-holomorphic regularization to be defined, and reproduces Brown's quasi-periods [2]. For properly meromorphic modular forms with poles inside the interior of $\mathbb{H}$, our L-function depends on the homotopy classes, defined in terms of pole-crossings. There is a natural choice of trivial homotopy class, which renders our L-functions canonical when the modular form has a pole in the interior of the fundamental domain, generalizing [3].

1 Introduction and preliminary concepts

Our primary goal in this note is to put forth a unifying generalization of the L-function associated to a given modular form. Our L-function naturally encompasses the classical case of L-functions associated to holomorphic modular forms [4], and seamlessly unifies Ref. [1]'s extension to L-functions of modular forms with poles at the cusp with Ref. [3]'s extension to L-functions of modular forms with poles at finite locations, exclusively away from cusps. Secondly, we show that our L-functions reproduce the numerical results for quasi-periods associated to weak cusp forms [2].

Note: Throughout this document, we refer to L-integrals and L-functions. The classical L-function-as-Dirichlet-series which has been suitably extended to the complex s plane is simply not defined when the L-function in question has poles either at the cusp or at finite locations within the upper half-plane. So, if there is ever a doubt, we state clearly: *any L-function in this note is defined by an L-integral*. The focus of this note is to define L-integrals for any integer weight $\mathrm{SL}_2(\mathbb{Z})$ modular form which may have poles at cusps or away from cusps.

1.1 Modular forms

Write $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$ for the real and imaginary parts of a complex number $z \in \mathbb{C}$. Let $\mathbb{H} := \{\tau \in \mathbb{C} \mid \mathrm{Im}(\tau) > 0\}$ denote the upper half-plane. $\mathrm{SL}_2(\mathbb{Z})$ is the group generated by $T := \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $S := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

There is a natural $\mathrm{SL}_2(\mathbb{Z})$ action on functions $f : \mathbb{H} \rightarrow \mathbb{C}$, graded by integers k : $(f|_k \gamma)(\tau) := (c\tau + d)^{-k} f(\gamma \cdot \tau)$ for all $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$. Similarly, $\gamma \cdot \tau := \frac{a\tau + b}{c\tau + d}$.

In this vein, a *holomorphic modular form* of weight k for $\mathrm{SL}_2(\mathbb{Z})$ is a holomorphic function $f : \mathbb{H} \rightarrow \mathbb{C}$ such that $(f|_k \gamma)(\tau) = f(\tau)$ for all $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ (aka $f(\tau + 1) = f(-\frac{1}{\tau})\tau^{-k} = f(\tau)$) for all

$\tau \in \mathbb{H}$, and $f(\tau) = \mathcal{O}(1)$ as $\text{Im}(\tau) \rightarrow \infty$, and which has no poles at finite locations within $\mathbb{H}$. By virtue of the periodicity $f(\tau) = f(\tau + 1)$ and holomorphicity, each holomorphic modular form has a q -series decomposition,

$$f(\tau) = \sum_n c_f(n) e^{2\pi i n \tau} = \sum_n c_f(n) q^n \quad , \quad q := e^{2\pi i \tau} \quad . \quad (1)$$

In order for f to be bounded as $\tau \rightarrow i\infty$, we conclude that $c_f(n) = 0$ for all $n < 0$, i.e. that $f = \sum_{n \geq 0} c_f(n) q^n$; for such modular forms f we write $f \in M_k$. If a modular form f additionally has a vanishing constant term, i.e. if $c_f(0) = 0$, then f is a *cusp form*, and we write $f \in S_k$. Clearly, $S_k \subset M_k$. Further, M_k is a finite-dimensional vector-space, as is S_k . Holomorphic $\text{SL}_2(\mathbb{Z})$ modular forms must have even integer weights, and $k \geq 4$ [4].

A product of modular forms of weights k_1 and k_2 is a modular form of weight $k_1 + k_2$, so $M_* := \bigoplus_k M_k$ is naturally a (graded) ring. This graded ring is freely generated by the two modular forms $E_4 \in M_4$ and $E_6 \in M_6$, the $k = 4, 6$ specializations of the *weight k Eisenstein series* E_k , which may be defined as follows:

$$E_k := \frac{1}{2} \sum_{(m,n)=1} \frac{1}{(m+n\tau)^k} = 1 - \frac{2k}{B_k} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n \quad , \quad (2)$$

where $\sum_{(m,n)=1}$ denotes summing over all relatively prime $(m, n) \in \mathbb{Z}^2$, the Bernoulli number B_n is the coefficient of $t^n/n!$ of the Laurent expansion of $t/(e^t - 1)$ about $t = 0$, and $\sigma_m(n) := \sum_{d|n} d^m$ is the Divisor function where $d|n$ means $n/d \in \mathbb{N}$. In short, $M_* = \mathbb{C}[E_4, E_6]$. We now define the *modular discriminant*:

$$\Delta := \frac{E_4^3 - E_6^2}{1728} = q + O(q^2) \in S_{12} \quad . \quad (3)$$

Holomorphicity is most canonically relaxed via allowing a modular form to have a pole at the cusp of $\text{SL}_2(\mathbb{Z})$, i.e. for a weight k modular form for $\text{SL}_2(\mathbb{Z})$ to have negative powers of q in its q -series expansion. In this case, f diverges as $\tau \rightarrow i\infty$ as $f \sim e^{-2\pi N \tau}$, where $-N$ is the largest negative number such that $c_f(n)$ is non-zero. Such a modular form f is called a *weak modular form*, and we write $f \in M_k^!$. If a weak modular form f has a vanishing coefficient of q^0 , i.e. if $c_f(0) = 0$, it is called a *weak cusp form* and we write $f \in S_k^!$. Clearly, $S_k^! \subset M_k^!$, $S_k \subset S_k^!$ and $M_k \subset M_k^!$. Two important weak cusp forms in our paper are as follows:

$$J := \frac{E_4^3}{\Delta} - 744 = \frac{1}{q} + O(q) \in S_0^! \quad , \quad (4)$$

$$\widehat{\Delta} = (J^2 + 24J - 393444)\Delta := \frac{1}{q} + O(q^2) \in S_{12}^! \quad . \quad (5)$$

In this work, we further relax from even weakly holomorphic modular forms, and consider *meromorphic modular forms* of weight k which are quotients: $f = \frac{f_1}{f_2}$ where $f_i \in M_{k_i}$, the denominator f_2 is not identically zero, and $k = k_1 - k_2$. We write F_k for the vector space of meromorphic modular forms of weight k , and set $F_* := \bigoplus_k F_k$. Then F_k vanishes unless k is an even integer, and F_* may be regarded as a “graded field of fractions” of M_* .

1.2 L-functions for holomorphic modular forms and periods for cusp forms

For any $f \in S_k$, one may define the associated L-function:

$$L(f, s) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s) \stackrel{f \in S_k}{:=} \frac{(2\pi)^s}{\Gamma(s)} \int_0^\infty f(it) t^{s-1} dt = \sum_{n \geq 1} \frac{c_f(n)}{n^s} \quad . \quad (6)$$

A similar definition applies for $f \in M_k$. (The classical definition of $L^*(f, s)$ for $f \in M_k$ is the focus of Remark 3.6, below.) Similarly, one may define a period polynomial for $f \in S_k$ in terms of this L-function,

$$r_f(X, Y) := (2\pi i)^{k-1} \sum_{\ell=0}^{k-2} (-1)^\ell \binom{k-2}{\ell} i^{\ell+1} L^*(f, \ell+1) X^{k-2-\ell} Y^\ell. \quad (7)$$

This polynomial, $r_f(X, Y) \in V_{k-2} := \mathbb{C}[X, Y]_{k-2}$, has a natural action under $\mathrm{SL}_2(\mathbb{Z})$, $(P|_\gamma)(X, Y) = P(aX + bY, cX + dY)$ for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$. This period polynomial can be decomposed into a basis of highly constrained polynomials, which are annihilated by $|(1+T)$ and $|(1+U+U^2)[6]$, where $U := TS$ generates an order-3 cyclic subgroup of $\mathrm{SL}_2(\mathbb{Z})$. We define the *periods* of a cusp form to be coefficients of r_f within this highly constrained sub-space W of V_{k-2} , i.e. $W \subset V_{k-2}$.

Why this paper? As classically computed, these L-functions and associated period polynomials (when defined in terms of the special values of the associated L-function) are not defined when f has poles either at the cusp of $\mathrm{SL}_2(\mathbb{Z})$ or within the interior of $\mathbb{H}$. As stated above, there are extant, but partial and non-overlapping, extensions of the definition of $L(f, s)$ to $f \in M_k^!$ [1] or $f \in F_k$ when f is regular at the cusp of $\mathrm{SL}_2(\mathbb{Z})$ [3]. In this note, we provide a construction that unifies these disparate extensions. We start with generic L-integrals in section 2, then specialize to $f \in M_k^!$ in section 3, and finally explore the more general context of $f \in F_k$ in section 4.

2 A finite L-integral

Definition 2.1. Let $\tau_0 \in \mathbb{H}$ and $f \in F_k$, let $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ denote the generators of $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, and let $f^{-1}(\infty)$ be the locus of all poles of $f \in \mathbb{H}$ (inclusive of the cusp of $\mathrm{SL}_2(\mathbb{Z})$). The S - and T -segments at τ_0 are smooth paths in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ from $\sigma^{-1}\tau_0$ to τ_0 ,

$$\gamma_{\tau_0}^S : -1/\tau_0 \longrightarrow \tau_0, \quad \gamma_{\tau_0}^T : \tau_0 - 1 \longrightarrow \tau_0. \quad (8)$$

Avoiding poles is automatic for $f \in M_k^!$, as long as the base-point τ_0 , is not at a cusp of $\mathrm{SL}_2(\mathbb{Z})$, i.e. there is only the trivial homotopy class. When $f \in F_k$, and f has poles at finite and non-zero imaginary heights, integration contours fall into nontrivial homotopy classes.

Definition 2.2. Let $\zeta(s, x) = \sum_{m \geq 0} (m+x)^{-s}$ as a function $\zeta : \mathbb{C}_s \times (\mathbb{C} \setminus \mathbb{N})_x \rightarrow \mathbb{C}$ denote the Hurwitz zeta function (meromorphic in s , holomorphic in $x \in \mathbb{H}$). We define Hurwitz T -kernel as

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(1-(k-s), \tau+1). \quad (9)$$

Definition 2.3 (Generic L-integral). Let $f \in F_k$, $\tau_0 \in \mathbb{H}$, and $s \in \mathbb{C}$. Further, let $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ be the S - and T -segments at τ_0 , as in Definition 2.1, and let k_T be as defined in Definition 2.2. The L-integral of f is

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := L_S^*(f, s; \gamma_{\tau_0}^S) + L_T^*(f, s; \gamma_{\tau_0}^T) := e^{-\pi i s/2} \left(\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right) \quad (10)$$

We use this L -integral to define a generalized L -function:

Definition 2.4 (Generic L-function). Let $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ be as defined in Definition 2.3. We define the generalized L-function of $f \in F_k$ to be,

$$L(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T). \quad (11)$$

Having defined the L-integral and L-function, we prove they are manifestly finite for $f \in F_k$:

Lemma 2.5 (Finiteness of L-integrals). *For $f \in F_k$, both $L_S^*(f, s; \gamma_{\tau_0}^S)$ and $L_T^*(f, s; \gamma_{\tau_0}^T)$ in Definition 2.3 are each, separately finite for every $s \in \mathbb{C}$, every $\tau_0 \in \mathbb{H}$, and for every path $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$.*

Proof. Both segments are compact and lie in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, so f is continuous on them; τ^{s-1} is continuous on $\mathbb{H}$, and $\tilde{k}_T(\tau, s)$ is continuous there since $\zeta(\nu, \tau + 1)$ is holomorphic in $\tau \in \mathbb{H}$ (Definition 2.2). A continuous integrand on a compact path is integrable, for every $s \in \mathbb{C}$. $\square$

We now turn to path-dependence and τ_0 -independence of the L-integral and the L-function of $f \in F_k$. This holds for all $f \in M_k^!$:

Proposition 2.6 (Basepoint & path independence for $f \in M_k^!$). *For $f \in M_k^!$ the value $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is independent of $\tau_0 \in \mathbb{H}$ and of the paths: with no interior poles, $\mathbb{H}$ is simply connected, so all admissible contours are homotopic. We write $L^*(f, s)$ for the common value and $L(f, s) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s)$.*

Proof. With no interior poles the integrands $f(\tau)\tau^{s-1}$ and $f(\tau)\tilde{k}_T(\tau, s)$ are holomorphic on $\mathbb{H}$, which is simply connected; by Cauchy's theorem each integral is unchanged under any deformation fixing its endpoints, hence depends on those endpoints alone. This is path-independence.

For independence of τ_0 show that variations in τ_0 for the S-term and T-term in $L^*(f, s)$ cancel, i.e. that $\partial_{\tau_0} L_S^*(f, s; \gamma_{\tau_0}^S) = -\partial_{\tau_0} L_T^*(f, s; \gamma_{\tau_0}^T)$. For L_T^* , note that the T -segment runs from $T^{-1}\tau_0 = \tau_0 - 1$ to τ_0 — two endpoints a unit apart, moving together with τ_0 — so by Leibniz's rule and the T -periodicity $f(\tau_0 - 1) = f(\tau_0)$,

$$\partial_{\tau_0} L_T^*(f, s; \gamma_{\tau_0}^T) = f(\tau_0) \tilde{k}_T(\tau_0, s) - f(\tau_0 - 1) \tilde{k}_T(\tau_0 - 1, s) = f(\tau_0) \delta_T \tilde{k}_T(\tau_0 - 1, s), \quad (12)$$

where $\delta_T \tilde{k}_T(\tau, s) := \tilde{k}_T(\tau + 1, s) - \tilde{k}_T(\tau, s)$. This is the whole role of δ_T : the unit gap between the endpoints turns the ∂_{τ_0} boundary term into a unit shift of the kernel. That shift is elementary — the kernel was built for it — since $\zeta(\nu, x + 1) - \zeta(\nu, x) = -x^{-\nu}$ gives, on both terms at $x = \tau + 1$,

$$\delta_T \tilde{k}_T(\tau, s) = -(\tau + 1)^{s-1} + e^{i\pi(s-1)}(\tau + 1)^{k-1-s} \quad (13)$$

$$\implies \partial_{\tau_0} L_T^*(f, s; \gamma_{\tau_0}^T) = f(\tau_0) [-\tau_0^{s-1} + e^{i\pi(s-1)}\tau_0^{k-1-s}]. \quad (14)$$

The S -segment runs $-1/\tau_0 \rightarrow \tau_0$; its lower endpoint moves with $\partial_{\tau_0}(-1/\tau_0) = \tau_0^{-2}$, so by Leibniz, the modularity $f(-1/\tau_0) = \tau_0^k f(\tau_0)$, and the phase $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)}\tau_0^{1-s}$,

$$\partial_{\tau_0} L_S^*(f, s; \gamma_{\tau_0}^S) = f(\tau_0)\tau_0^{s-1} - f(-1/\tau_0)(-1/\tau_0)^{s-1}\tau_0^{-2} = f(\tau_0)[\tau_0^{s-1} - e^{i\pi(s-1)}\tau_0^{k-1-s}]. \quad (15)$$

Thus τ_0 -variations are equal and opposite, $\partial_{\tau_0} L_S^* = -\partial_{\tau_0} L_T^*$. Thus $\partial_{\tau_0} L^* = \partial_{\tau_0} L_S^* + \partial_{\tau_0} L_T^* = 0$. $\square$

Remark 2.7 (Recovering the classical L-integral for $f \in S_k$). *Let $f \in S_k$. As $f^{-1}(\infty)$ does not include the cusp $i\infty$, $\lim_{\tau_0 \rightarrow i\infty} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ exists. In this limit, the S -segment maps onto the standard L -function contour, $[0, i\infty]$. Simultaneously, as $f(it) \sim e^{-2\pi t}$ as $t \rightarrow \infty$, the T -segment vanishes exponentially. Thus, in this limit, $L^*(f, s)$ exactly maps onto the classical L -function integral [4]:*

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s) = \int_0^\infty t^{s-1} f(it) dt. \quad (16)$$

3 L-functions for weak modular forms

Proposition 3.1. Let $\Gamma(s, x) := \int_x^\infty e^{-t} t^{s-1} dt$ denote the incomplete Gamma function. For $n \neq 0$, $\nu \in \mathbb{C}$, and $z \in \mathbb{H}$,

$$\int_z^{z+1} e^{2\pi i n u} \zeta(\nu, u) du = e^{i\pi(1-\nu)/2} (2\pi n)^{\nu-1} \Gamma(1-\nu, -2\pi i n z). \quad (17)$$

Proof. Write $\zeta(\nu, u) = \sum_{m \geq 0} (m+u)^{-\nu}$ and shift $u \mapsto u+m$ in the m -th term; as $e^{2\pi i n(u+m)} = e^{2\pi i n u}$ for integer m , the unit segments stack into one horizontal ray,

$$\int_z^{z+1} e^{2\pi i n u} \zeta(\nu, u) du = \sum_{m \geq 0} \int_{z+m}^{z+1+m} e^{2\pi i n v} v^{-\nu} dv = \int_z^{z+\infty} e^{2\pi i n v} v^{-\nu} dv. \quad (18)$$

Substitute $w = -2\pi i n v$, so that $v = iw/(2\pi n)$, $dv = i dw/(2\pi n)$, and $e^{2\pi i n v} = e^{-w}$: the lower endpoint becomes $w = -2\pi i n z$, and for $n > 0$ the ray rotates to $w \rightarrow +\infty$ along the positive reals — legal, since $e^{-w} w^{-\nu}$ decays in the right half-plane, so the closing arc vanishes. With $v^{-\nu} = i^{-\nu} (2\pi n)^{\nu-\nu} w^{-\nu}$,

$$\int_z^{z+\infty} e^{2\pi i n v} v^{-\nu} dv = i^{1-\nu} (2\pi n)^{\nu-1} \int_{-2\pi i n z}^{\infty} e^{-w} w^{-\nu} dw = e^{i\pi(1-\nu)/2} (2\pi n)^{\nu-1} \Gamma(1-\nu, -2\pi i n z). \quad (19)$$

For $n < 0$ the ray instead grows as $\text{Im} v \rightarrow \infty$ and the rotation is illegal; but both sides are analytic in n and agree for $n > 0$, so the identity continues to $n < 0$ with Γ on its principal branch. At $n = 0$ the prefactor $(2\pi n)^{\nu-1}$ is singular; that case is handled separately. $\square$

Theorem 3.2 (The L-function for $f \in M_k^!$). For $f \in M_k^!$, we may set $\tau_0 = i$, and prove

$$L^*(f, s) = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left(\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right). \quad (20)$$

Here, trivially, $L^*(f, s) = i^k L^*(f, k-s)$. The L-function is $L(f, s) := (2\pi)^s L^*(f, s)/\Gamma(s)$.

Proof. By path independence and base-point independence from Proposition 2.6, we may freely set $\tau_0 = i$. Here, $L_S^*(f, s; \gamma_{\tau_0}^S) = 0$ (the S -contour vanishes in length and there are no nontrivial homotopy classes as $f \in M_k^!$ has no poles which a finite contour can wrap), and the L-integral simplifies dramatically:

$$L^*(f, s) = L_T^*(f, s; \gamma_i^T) = e^{-\pi i s/2} \int_{i-1}^i f(\tau) \tilde{k}_T(\tau, s) d\tau. \quad (21)$$

We then expand f in terms of its q-series, $\sum_n c_f(n) e^{2\pi i n \tau}$, and substitute the u -variable of Proposition 3.1 as $u = \tau + 1$, and note that $e^{2\pi i n(u-1)} = e^{2\pi i n u}$. When $n \neq 0$, straightforward substitutions of ν, z into Proposition 3.1 suffice: $1-\nu = s$ gives the contribution of the first term in $\tilde{k}_T(\tau, s)$, and $1-\nu = k-s$ gives the second:

$$e^{-\pi i s/2} \left(e^{\pi i s/2} \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} - e^{i\pi(s-1)} e^{\pi i(k-s)/2} \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right) = \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}. \quad (22)$$

We may then evaluate the $n = 0$ mode's contribution follows from noting the closed form solution for the definite integral,

$$\int_{i-1}^i \zeta(\sigma, \tau + 1) d\tau = \int_i^{i+1} \zeta(\sigma, u) du = \frac{i^{1-\sigma}}{\sigma-1}. \quad (23)$$

This result follows from noting $\partial_a(\zeta(\sigma-1, a)/(1-\sigma)) = \zeta(\sigma, a)$ and $\zeta(\sigma-1, a+1) - \zeta(\sigma-1, a) = -a^{1-\sigma}$. Applying this to the two distinct terms of $\tilde{k}_T(\tau, s)$ at $\sigma = 1-s$ and $\sigma = 1-(k-s)$ gives

$$e^{-\pi is/2} \int_{i-1}^i c_f(0) \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right). \quad (24)$$

Combining the $n=0$ contributions with the $n \neq 0$ contributions reproduces Eq. (20). Finally, the functional equation, $L^*(f, s) = i^k L^*(f, k-s)$, follows by direct inspection. $\square$

3.1 Explicit S- and T-contours for general τ_0

Lemma 3.3 (L_T^* for $M_k^!$). *For $f \in M_k^!$, and $\tau_0 \in \mathbb{H}$ away from cusps, $L_T^*(f, s; \gamma_{\tau_0}^T)$ evaluates to:*

$$L_T^*(f, s; \gamma_{\tau_0}^T) = -c_f(0) \left(\frac{(\tau_0/i)^s}{s} + i^k \frac{(\tau_0/i)^{k-s}}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left(\frac{\Gamma(s, -2\pi in\tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, -2\pi in\tau_0)}{(2\pi n)^{k-s}} \right). \quad (25)$$

Proof. Expand $f = \sum_n c_f(n) q^n$ on the compact segment $[\tau_0-1, \tau_0]$, and restrict attention to the sub-sum, $\sum_{n \neq 0}$. For each such $n \neq 0$, apply the Proposition 3.1 in $\tilde{k}_T(\tau, s) = \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1)$ at $\nu = 1-s$ and $\nu = s-(k-1)$. The phases for the second term work-out identically as in the proof of Theorem 3.2.

For $n=0$, the constant $c_f(0)$ multiplies $e^{-\pi is/2} \int_{\tau_0-1}^{\tau_0} \tilde{k}_T(\tau, s) d\tau$; the Hurwitz antiderivative $\int_{\tau_0-1}^{\tau_0} \zeta(\sigma, \tau+1) d\tau = \tau_0^{1-\sigma}/(\sigma-1) -$ from $\partial_\tau \zeta(\sigma-1, \tau+1) = -(\sigma-1)\zeta(\sigma, \tau+1)$ and the shift $\zeta(\sigma-1, \tau_0+1) - \zeta(\sigma-1, \tau_0) = -\tau_0^{1-\sigma}$ — applied to the two halves of $\tilde{k}_T$ at $\sigma = 1-s$ and $\sigma = s-k+1$ yields $-e^{-\pi is/2} \tau_0^s/s = -(\tau_0/i)^s/s$ and $-e^{\pi is/2} \tau_0^{k-s}/(k-s) = -i^k (\tau_0/i)^{k-s}/(k-s)$ respectively. $\square$

Corollary 3.4 (L_T^* as $\tau_0 \rightarrow i\infty$). *Let $f \in S_k^!$, where $f = \sum_{n \geq -N} c_f(n) q^n$ for some $-N \in \mathbb{Z}$. The T-contour's contribution to $L^*(f, s)$ is given by Lemma 3.3. As $\tau_0 = it$ goes to the cusp $i\infty$, the T-contour scales exactly as $f(it)$ scales: $L_T^*(f, s; \gamma_{\tau_0}^T) \sim f(it) \sim e^{2\pi Nt}$.*

Proof. For the asymptotics, $\Gamma(\sigma, z) \sim z^{\sigma-1} e^{-z}$ as $|z| \rightarrow \infty$ for $|\arg z| < \frac{3\pi}{2}$ [10, §8.11] gives $|c_f(n)|$ -weighted go as $e^{-2\pi n \text{Im}(\tau_0)}$. The smallest mode n in the q-series, i.e. $n = -N$, dominates the scaling of $f(it)$ and of $L_T^*(f, s; \gamma_{it}^T)$. Combining these results yields the claimed asymptotics. $\square$

Now, exploiting τ_0 -independence of $L^*(f, s)$, its explicit value as given in Theorem 3.2, and the explicit T-contribution $L_T^*(f, s; \gamma_{\tau_0}^T)$ trivially implies clean, closed-form expressions for $L_S^*(f, s; \gamma_{\tau_0}^S)$:

Corollary 3.5 (L_S^* for $M_k^!$). *For $f \in M_k^!$ and $\tau_0 \in \mathbb{H}$, $L_S^*(f, s; \gamma_{\tau_0}^S)$ evaluates to*

$$L_S^*(f, s; \gamma_{\tau_0}^S) = c_f(0) \left(\frac{(\tau_0/i)^s - 1}{s} + i^k \frac{(\tau_0/i)^{k-s} - 1}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left(\frac{\Gamma(s, 2\pi n) - \Gamma(s, -2\pi in\tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n) - \Gamma(k-s, -2\pi in\tau_0)}{(2\pi n)^{k-s}} \right). \quad (26)$$

Proof. Recall $L^*(f, s) - L_T^*(f, s; \gamma_{\tau_0}^T) = L_S^*(f, s; \gamma_{\tau_0}^S)$. Now, $L^*(f, s)$ for $f \in M_k^!$ is given by Theorem 3.2, and $L_T^*(f, s; \gamma_{\tau_0}^T)$ is given by Lemma 3.3. Their difference yields Eq. (26). $\square$

3.2 Classical Mellin recovery and canonical uniqueness.

As discussed in Remark 2.7, for $f \in S_k$, taking the cusp-limit $\tau_0 \rightarrow i\infty$ in Theorem 3.2 collapses the S -segment onto the classical Mellin contour $[0, i\infty]$ while via Corollary 3.4 the T -segment vanishes exponentially as $L_T^*(f, s; \gamma_{\tau_0}^T) \sim f(\tau_0) \sim e^{-2\pi\text{Im}(\tau_0)N}$ (as $f \in S_k \Leftrightarrow f = \sum_{n \geq N} c_f(n)q^n$ with $N \geq 1$). I.e., here, one recovers the classical L-integral of f without obstruction:

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = \int_0^\infty t^{s-1} f(it) dt. \quad (27)$$

This story changes when $f = 1/q^N(1 + O(q)) \in S_k^!$ for some $N \geq 1$.

In short, when f has a polar q -series, it has a pole at the cusp of $\text{SL}_2(\mathbb{Z})$. Thus, the limit $\tau_0 \rightarrow i\infty$, which was crucial to recovering the classical Mellin contour in Remark 2.7 simply does not exist.

However, it can still be instructive to examine $L_T^*(f, s; \gamma_{\tau_0}^T)$, and $L_S^*(f, s; \gamma_{\tau_0}^S)$ as τ_0 goes towards $i\infty$. In this context, while of course the sum $L_T^*(f, s; \gamma_{\tau_0}^T) + L_S^*(f, s; \gamma_{\tau_0}^S) = L^*(f, s)$ is manifestly invariant under changes of the base-point, each of these two terms can indeed change values. Indeed, Corollary 3.4 clearly shows that as $\tau_0 \rightarrow i\infty$, when $f \in S_k^!$ has negative powers of q in its q -series, then $L_T^*(f, s; \gamma_{\tau_0}^T) \sim e^{2\pi N \text{Im}(\tau_0)}$. Discarding $\gamma_{\tau_0}^T$ therefore does violence to the result: When the S -contour is isolated from its T -partner, and $\tau_0 \rightarrow i\infty$, it diverges. This is the precise reason that [1] needed to regularize their divergent L-integral representation for properly weak cusp forms.

That said, there is one further aspect of our L-function even in the classical case of $f \in M_k$ that is worthy of remark:

Remark 3.6 (Recovering the classical L-integral for $f \in M_k$). *In textbook treatments [4], when $f \in M_k$, its L-integral is defined as*

$$\begin{aligned} L^*(f, s)|_{\text{cl}} &:= \int_0^\infty (f(it) - c_f(0))t^{s-1}dt = \frac{\Gamma(s)}{(2\pi)^s} \sum_{n=1}^\infty \frac{c_f(n)}{n^s} \\ &= -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n=1}^\infty c_f(n) \left(\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right) \end{aligned} \quad (28)$$

Classically, this constant-term subtraction is needed because $\int_0^\infty t^{s-1}dt$ is a pure divergence, so the q -series cannot commute with the integration contour.

However, as is clear from our general Theorem 3.2 for L-integrals for $f \in M_k^!$, and from the specific results of the S - and T -contours in Lemma 3.3 and Corollary 3.5, our L-integral requires no such explicit subtraction of the constant term $-c_f(0)$. Thus, not only does our prescription naturally handle divergences at poles when $f \in M_k^!$ has polar terms in its q -series, but it also obviates the need to by-hand remove the constant mode from the kernel of the L-integral that is needed when defining L-functions in terms of the classical Mellin contour, $\int_0^\infty$.

3.3 Period polynomials, periods and quasi-periods

Because $L^*(f, s)$ is canonical for every $f \in S_k^!$, and because period polynomials for a given modular form can be written in terms of their L-functions evaluated at integer- s in the critical strip, they need no separate construction.

Definition 3.7 (Period polynomial for $S_k^!$). *For $f \in S_k^!$ and $n = k - 2$, the period polynomial is*

$$r_f(X, Y) := (2\pi i)^{n+1} \sum_{\ell=0}^n (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1) X^{n-\ell} Y^\ell. \quad (29)$$

Proposition 3.8 (Period polynomial as a contour integral). *Let $n := k - 2$. With $(X - \tau Y)^n$ in place of the L -weight τ^{s-1} , and $\mathbf{K}_T(\tau; X, Y) := \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell \tilde{k}_T(\tau, \ell + 1) X^{n-\ell} Y^\ell$ in place of $\tilde{k}_T$, the same two segments assemble r_f directly:*

$$r_f(X, Y) = (2\pi i)^{n+1} \left[\int_{\gamma_{\tau_0}^S} f(\tau) (X - \tau Y)^n d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \mathbf{K}_T(\tau; X, Y) d\tau \right]. \quad (30)$$

Proof. By Definition 2.3, the $X^{n-\ell} Y^\ell$ coefficient of the bracket is

$$r_f|_{X^{n-\ell} Y^\ell} = (-1)^\ell \binom{n}{\ell} i^{\ell+1} [L_S^*(f, \ell + 1) + L_T^*(f, \ell + 1)] = (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell + 1). \quad (31)$$

This matches Definition 3.7 term by term, and proves the claim. $\square$

Having now defined the period polynomial associated to a weak cusp form $f \in S_k^!$, we now seek to find its (*quasi*-)periods. First, a number of helpful definitions. For starters, let $f \in S_k^!$, $\tau_0 \in \mathbb{H}$ and $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, and define

$$C_\gamma(f; \tau_0) := (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} \omega_f, \quad \text{where } \omega_f := f(\tau)(X - \tau Y)^n d\tau. \quad (32)$$

Finally, it is helpful to define two slight abstractions related to $\mathbf{K}_T(\tau; X, Y)$ and to $L_T^*(f, s; \gamma_{\tau_0}^T)$:

$$\chi(\tau; X, Y) := \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell \zeta(-\ell, \tau + 1) X^{n-\ell} Y^\ell, \quad (33)$$

$$E(f; \tau_0) := (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau) \chi(\tau; X, Y) d\tau. \quad (34)$$

For each τ , $\chi(\tau; \cdot, \cdot) \in V_{k-2} = \mathbb{C}[X, Y]_{k-2}$; the $\mathrm{SL}_2(\mathbb{Z})$ -slash $(P|_\gamma)(X, Y) := P(aX + bY, cX + dY)$ acts on the polynomial variables alone, holding τ fixed:

$$\chi|_\gamma(\tau; X, Y) := \chi(\tau; aX + bY, cX + dY), \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}). \quad (35)$$

Thus $|_T$ sends $(X, Y) \mapsto (X + Y, Y)$ with τ untouched. Being linear and τ -independent, $|_\gamma$ commutes with $\int d\tau$, so slashing the polynomial E agrees with slashing χ under the integral — the two occurrences of $|_\gamma$ in Lemma 3.9 denote the same operation.

Lemma 3.9. *Let $f \in S_k^!$, $\tau_0 \in \mathbb{H}$, and let ω_f , $C_\gamma(f; \tau_0)$, $\chi(\tau; X, Y)$, and $E(f; \tau_0)$ be as defined in Eqs. (32), (33) and (34). Then $\hat{C}_\gamma := C_\gamma - (E|_\gamma - E)$ is a 1-cocycle for $\mathrm{SL}_2(\mathbb{Z})$ with $\hat{C}_S = r_f$ and $\hat{C}_T = 0$.*

Proof. First, we prove $\hat{C}_\gamma$ is a cocycle. Path-additivity and covariance of ω_f give the Eichler–Shimura cocycle relation $C_{\gamma\delta} = C_\gamma|_\delta + C_\delta$ [11, 6]; as $\gamma \mapsto E|_\gamma - E$ is a coboundary, $\hat{C}$ is a cocycle.

Having established it is a cocycle, we now show that $\hat{C}_S = r_f$. At $s = \ell + 1$, Definition 2.2 reads $\tilde{k}_T(\tau, \ell + 1) = \zeta(-\ell, \tau + 1) - (-1)^\ell \zeta(-(n - \ell), \tau + 1)$; assembling $\mathbf{K}_T$ and re-indexing $\ell \mapsto n - \ell$ in the second term (with n even) gives $\mathbf{K}_T = \chi - \chi|_S$. So the T -term of r_f is $E - E|_S$, and by Proposition 3.8 $r_f = C_S + (E - E|_S) = C_S - (E|_S - E) = \hat{C}_S$.

Finally, we show that $\hat{C}_T = 0$. The Hurwitz shift $\zeta(-\ell, \tau) = \zeta(-\ell, \tau + 1) + \tau^\ell$ gives the point-wise identity $\chi(\tau - 1; X, Y) - \chi(\tau; X, Y) = (X - \tau Y)^n$; applied twice it shows $A(\tau) :=$

$\chi(\tau; X+Y, Y) - \chi(\tau-1; X, Y)$ satisfies $A(\tau-1) = A(\tau)$. Crucially, because $A(\tau)$ is a polynomial, and because $A(\tau) = A(\tau+1)$ for generic τ , then $A(\tau)$ is constant: $\partial_\tau A(\tau) = 0 \Leftrightarrow A(\tau) = A_0$. Thus,

$$\begin{aligned}
\hat{C}_T &= C_T - (E|_T - E) \\
&= (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau)(X - \tau Y)^n d\tau - (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau)[\chi|_T(\tau; X, Y) - \chi(\tau; X, Y)] d\tau \\
&= (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau) \left([\chi(\tau-1; X, Y) - \chi(\tau; X, Y)] - [\chi(\tau; X+Y, Y) - \chi(\tau; X, Y)] \right) d\tau \\
&= (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau) [\chi(\tau-1; X, Y) - \chi(\tau; X+Y, Y)] d\tau = -(2\pi i)^{n+1} \int_{\tau_0-1}^{\tau_0} f(\tau) A(\tau) d\tau \\
&= -(2\pi i)^{n+1} A_0 \int_{\tau_0-1}^{\tau_0} f(\tau) d\tau = -(2\pi i)^{n+1} A_0 c_f(0) = 0,
\end{aligned} \tag{36}$$

as all elements $f \in S_k^!$ must have $c_f(0) = 0$. $\square$

Definition 3.10. $W := \{w \in \mathbb{C}[X, Y]_{k-2} : w|_{(1+S)} = w|_{(1+U+U^2)} = 0\} \subset V_n$

Lemma 3.11. For $f \in S_k^!$, and $W \subset V_n$ from Definition 3.10, the period polynomial $r_f(X, Y) \in W$.

Proof. Let $\hat{C}_\gamma$ be the parabolic cocycle of Lemma 3.9. Thus, $r_f = \hat{C}_S$ and $\hat{C}_T = 0$. Now, note that $\hat{C}_\gamma(f; \tau_0) = 0$ when $\gamma = \pm I$ for the simple reason that when $\gamma = \pm I$, $\hat{C}_\gamma$ is a closed-contour integral of a function which has no poles in $\mathbb{H}$. Finally, direct computation, starting from the identities $-I = U^3 = S^2$ and using the cocycle law $\hat{C}_{\gamma\delta} = \hat{C}_\gamma|_\delta + \hat{C}_\delta$ and clever use of $\hat{C}_T = 0$, shows $r_f|_{(1+S)} = r_f|_{(1+U+U^2)} = 0$:

$$\hat{C}_U = \hat{C}_{TS} = \hat{C}_T|_S + \hat{C}_S = \hat{C}_S = r_f, \tag{37}$$

$$0 = \hat{C}_{S^2} = \hat{C}_S|_S + \hat{C}_S = \hat{C}_S|_{(1+S)} = r_f|_{(1+S)}, \tag{38}$$

$$0 = \hat{C}_{U^3} = \hat{C}_U|_{UU} + \hat{C}_{UU} = \hat{C}_U|_{U^2} + (\hat{C}_U|_U + \hat{C}_U) = \hat{C}_U|_{(1+U+U^2)} = r_f|_{(1+U+U^2)}. \tag{39}$$

Thus, $r_f \in W \subset V_n$, and finishes the proof. $\square$

To define periods and quasi-periods, we recall the standard fact that $\dim(W) = 2 \dim(S_k) + 1$ (see, e.g. [6]), and define the inner-product:

$$\langle P, Q \rangle := \sum_a (-1)^a \binom{n}{a}^{-1} p_a q_{n-a}. \tag{40}$$

Now for the basis elements for k where $\dim(S_k) = 1$:

Definition 3.12. Let $\dim S_k = 1$, $n := k-2$, $p_0(X) := X^n - 1$, $B_\ell := \frac{\ell!}{2\pi i} \oint \frac{t^{-\ell}}{e^t - 1} dt$, and $B_m^0(X) := \sum_{i \neq 1} \binom{m}{i} B_i X^{m-i}$. Motivated by [6, Thm. 1'], we define:

$$W_-^{(k)}(X) := \text{pp} \left(\frac{1}{n-1} (B_{n-1}^0(X) + X^n B_{n-1}^0(\frac{1}{X})) - \frac{1}{3} (B_3^0(X) + X^n B_3^0(\frac{1}{X})) \right), \tag{41}$$

$$W_+^{(k)}(X) := \text{pp} \left(\frac{1}{2} (B_2^0(X) - X^n B_2^0(\frac{1}{X})) + \frac{1}{n} (B_n^0(X) - X^n B_n^0(\frac{1}{X})) - c p_0(X) \right), \tag{42}$$

$$v_0^{(k)}(X) := p_0(X) - W_+^{(k)}(X) \left(\langle W_+^{(k)}, p_0 \rangle / \langle W_+^{(k)}, W_+^{(k)} \rangle \right), \tag{43}$$

where pp acts on any polynomial with rational coefficients by multiplying by all coefficient-denominators and dividing by the gcd, c eliminates the $p_0(X)$ component of $W_+^{(k)}$, and $\langle P, Q \rangle$ as in Eq. (40).

Lemma 3.13. *For k with $\dim S_k = 1$, the space W of Lemma 3.11 has $\dim W = 3$, and $\{W_+^{(k)}, W_-^{(k)}, v_0^{(k)}\}$ is an orthogonal basis of W under $\langle \cdot, \cdot \rangle$ from Eq. (40).*

Proof. First, let $W_\pm^{(k)}(X, Y) := Y^{k-2}W_\pm^{(k)}(X/Y)$, and let $v_0^{(k)}(X, Y) := Y^{k-2}v_0^{(k)}(X/Y)$.

Now, by Eichler–Shimura, $\dim W = 2 \dim S_k + 1 = 3$. Theorem 1' of [6] gives the odd and even period polynomials of the cusp form $R_n \propto \Delta$ as modified-Bernoulli combinations; its $n = 2$ case is $r^-(R_2)$, odd under $X \mapsto -X$, and its $n = 1$ case is $r^+(R_1)$, even. Our $W_-^{(k)}$ is the primitive part of $r^-(R_2)$, and $W_+^{(k)}$ the primitive part of $r^+(R_1)$ with its Eisenstein p_0 -component removed by c ; both lie in W , as does $p_0 = X^{k-2} - Y^{k-2}$ (even). The three are independent, hence a basis of W . Since $k - 2$ is even, $\langle \cdot, \cdot \rangle$ is symmetric and pairs opposite $X \mapsto -X$ parities to zero; the odd $W_-^{(k)}$ is thus orthogonal to the even $W_+^{(k)}$ and $v_0^{(k)}$, while $v_0^{(k)} \perp W_+^{(k)}$ by construction. Hence $\{W_+^{(k)}, W_-^{(k)}, v_0^{(k)}\}$ is orthogonal. $\square$

Definition 3.14 (Periods and Quasi-Periods). *Let k be such that $\dim(S_k) = 1$, $n := k - 2$, $W_\pm^{(k)}(X, Y) := Y^n W_\pm^{(k)}(X/Y)$ be as in Definition 3.12, and $r_f(X, Y)$ be as in Definition 3.7. Now, define the two numbers $\omega^\pm(f)$ as follows:*

$$\omega^\pm(f) := \frac{\langle r_f, W_\pm^{(k)} \rangle}{\langle W_\pm^{(k)}, W_\pm^{(k)} \rangle}, \quad (44)$$

where $\langle \cdot, \cdot \rangle$ is as in Eq. (40). When f is a cusp form, $\omega^\pm(f)$ are its periods; when f is a weak cusp form, $\omega^\pm(f)$ are its quasi-periods, sometimes denoted $\eta^\pm(f)$.

Now, having defined our periods and quasi-periods, we use them together with our L-functions to extract periods and quasi-periods for $f \in S_k^!$ (when $\dim(S_k) = 1$). To this effect, we recall the notation of [5]. Decomposing the modular weight $k = 12\lambda + k'$ where $k' \in \{0, 4, 6, 8, 10, 14\}$, we see $\lambda = \dim(S_k)$ when $\lambda \geq 0$. For any λ , we may denote $f_{k,m} := 1/q^m + O(q^{\lambda+1}) \in M_k^!$. When $\dim(S_k) = 1$, we have $f_{k,m} = 1/q^m + O(q^2)$. Here, define $\Delta_k := f_{k,-1} = q + O(q^2) \in S_k$ and $\hat{\Delta}_k := f_{k,1} = 1/q + O(q^2) \in S_k^!$. Further, define

$$\text{pd}_k := \begin{pmatrix} \omega^+(\Delta_k) & \omega^-(\Delta_k) \\ \eta^+(\hat{\Delta}_k) & \eta^-(\hat{\Delta}_k) \end{pmatrix}, \quad \alpha_k := \frac{(-1)^{1+k/2} \det(\text{pd}_k)}{(2\pi i)^{k-1} (k-2)!} \quad (45)$$

k	$\omega^+(\Delta_k)$	$\omega^-(\Delta_k)/i$	$\eta^+(\hat{\Delta}_k)$	$\eta^-(\hat{\Delta}_k)/i$	α_k
12	-6.89168e7	-5.58502e6	1.27202e11	1.02767e10	1
16	-8.81481e11	-1.54063e10	4.14120e13	6.83541e11	13/30
18	-5.07833e13	-5.43305e12	-2.19200e14	-3.36339e13	2/3
20	-4.22882e16	-3.73170e14	6.38640e16	-1.89823e14	17/5
22	-1.49974e18	-4.24489e16	1.43795e18	-4.41050e16	19/21
26	-1.18673e22	-3.60961e21	1.18653e22	-3.61016e21	23/15

Table 1: Periods $\omega^\pm(\Delta_k)$ and quasi-periods $\eta^\pm(\hat{\Delta}_k)$, and the rational number α_k . While $\alpha_{12} = 1$ matches Brown [2], α_k for $k \geq 16$ appear to be new, interesting, rational numbers, albeit with implicit dependence on the period-polynomial basis normalizations—in contrast to the normalization-free Petersson check above. We have not found a clear pattern for the α_k . See [9] for the scripts that extracted these figures (to 50-digit precision).

Brown–Hain [7, Prop. 5.6] show that $\det(\text{pd}_k)$ satisfies a Legendre-type relation, $\det(\text{pd}_k) \in \mathbb{Q}^\times \cdot (2\pi i)^{k-1}$. Brown [2, §8.3] numerically finds $\alpha_k = 1$; we reproduce this result at $k = 12$ and

give the analogous rational multiples for $k \in \{16, 18, 20, 22, 26\}$ in Table 1, below. Note: The value of α_k is tied to the normalization of Definition 3.12. Other normalizations, e.g. not including the pp-operator in defining $W_{\pm}^{(k)}$ would lead to different α_k -values.

4 Meromorphic modular forms

For $f \in F_k$ with poles in $\mathbb{H}$, the two segment integrals of Definition 2.3 depend on their contours; we show their sum depends only on the homotopy class of the pair.

Proposition 4.1. *For $f \in F_k$ and $\tau_0 \in \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, the L-integral $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ depends only on the homotopy class of its two segments in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, and not on the basepoint τ_0 . At fixed τ_0 each of L_S^*, L_T^* is separately path-independent.*

Proof. At fixed τ_0 the integrands $f(\tau)\tau^{s-1}$ and $f(\tau)\tilde{k}_T(\tau, s)$ are holomorphic on $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, so Cauchy's theorem makes each of L_S^*, L_T^* constant on the homotopy class of its own segment. Under a motion of the basepoint, $\partial_{\tau_0} L^*$ is evaluated at the moving endpoints alone; the computation of Proposition 2.6 — which uses only $f(\tau_0 - 1) = f(\tau_0)$ and $f(-1/\tau_0) = \tau_0^k f(\tau_0)$ — cancels the boundary terms of L_S^* against those of L_T^* , the interior poles lying off the contours. Hence $\partial_{\tau_0} L^* = 0$ along any motion in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, while each summand may separately vary. $\square$

Definition 4.2. *For $f \in F_k$ and $\tau_0 \in \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, let $[\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]$ denote the homotopy classes of the two segments in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$.*

Thus, with $\gamma_{\tau_0}^S \in [\gamma_{\tau_0}^S]$ and $\gamma_{\tau_0}^T \in [\gamma_{\tau_0}^T]$, by Proposition 4.1 the value

$$L^*(f, s; [\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]) := L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = L_S^*(f, s; \gamma_{\tau_0}^S) + L_T^*(f, s; \gamma_{\tau_0}^T) \quad (46)$$

is independent of the chosen representative paths and independent of τ_0 (subject to natural constraints of τ_0 monodromies around singular points, which move between homotopy classes).

We would like to speak of “the” L-integral. Homotopy classes of course split this into L-integrals for contours in different homotopy classes. And, so, in what follows, we define the reference homotopy classes:

Definition 4.3 (Reference homotopy class for $f \in F_k$). *Let $f \in F_k$, let τ_0 be the base-point of $L^*(f, s)$, and let $f^{-1}(\infty)$ be the poles of f . If f has no poles on the line $\text{Im}(\tau) = 1$, choose $\delta = 0$. Else, choose $\delta > 0$ such that no pole lies in the strip $1 < \text{Im}(\tau) \leq 1 + \delta$. Set $\tau_0^* := i(1 + \delta)$. The reference homotopy class $[\hat{\gamma}_f] = [\hat{\gamma}_f^T] + [\hat{\gamma}_f^S]$ is:*

- $[\hat{\gamma}_f^T]$: contains the horizontal segment from $T^{-1}\tau_0^*$ to τ_0^* ;
- $[\hat{\gamma}_f^S]$: contains the near-vertical path from $S^{-1}\tau_0^*$ to τ_0^* , which passes any pole at i to the left.

For $\delta = 0$, the reference paths match the paths $\gamma_i^S : i \rightarrow i$ and $\gamma_i^T : (i-1) \rightarrow i$ for $L^*(f, s)$ when $f \in M_k^1$.

With this definition, we may thus unambiguously write

$$L^*(f, s) := L^*(f, s; [\hat{\gamma}_f^S], [\hat{\gamma}_f^T]) . \quad (47)$$

Now, having fixed the reference value $L^*(f, s)$ in the reference homotopy class of Definition 4.3, consider two paths $\gamma_{\tau_0}^S$ and $\gamma_{\tau_0}^T$ which may be in any homotopy class. Further, let $X_S(p), X_T(p) \in \mathbb{Z}$ count the windings $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ has about each pole $\tau_p \in f^{-1}(\infty)$ relative to the reference-class.

Lemma 4.4. *Let $f \in F_k$, $r_S(f, \tau_p) := \text{Res}_{\tau=\tau_p}(f(\tau)\tau^{s-1})$ and $r_T(f, \tau_p) := \text{Res}_{\tau=\tau_p}(f(\tau)\tilde{k}_T(\tau, s))$. Then,*

$$L^*(f, s; [\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]) = L^*(f, s) + 2\pi i e^{-\pi i s/2} \sum_{\tau_p \in f^{-1}(\infty)} \left[X_S(\tau_p) r_S(f, \tau_p) + X_T(\tau_p) r_T(f, \tau_p) \right]. \quad (48)$$

Proof. Two pairs whose winding data differ by one unit at a single pole p , on one segment, bound a loop encircling p once; the residue theorem adds $2\pi i$ times the residue of that segment's integrand, $f(\tau)\tau^{s-1}$ for γ^S or $f(\tau)\tilde{k}_T(\tau, s)$ for γ^T , and the prefactor $e^{-\pi i s/2}$ of Definition 2.3 multiplies through. Summing over crossings gives (48). $\square$

This content provides the basics to discuss our finite L-functions for $f \in F_k$. However, we wish to present these L-functions in terms of their q-series coefficients, just as we have done for L-functions for $f \in M_k^!$ in the previous section. To do this, we must introduce polylogarithmic projections of Ref. [3]:

This polylog projection isolates the q-series contribution of a given pole of an $f \in F_k$ into a known special function, an integral of a polylogarithm of negative index, and leaves behind an object $(I - \hat{P})f$ with that one pole removed. For our purposes, we find it essential to project-out all poles at, and above, the line $\text{Im } \tau = 1$. We must do this because the reference contours which describe our reference homotopy classes are, at a minimum, contour integrals of f across the horizontal segment from $i - 1$ to i . Were we to allow poles at/above this line, q-series diverges on the contour ($|c_f(n)| \gtrsim n^{m-1} e^{2\pi n \text{Im } \tau_p}$ for a pole of order m [3]).

Within this general context, we lay-out the starting definitions and facts:

Definition 4.5. *Let $f \in F_k$ have, at a pole τ_p of order m_p , the Laurent principal part $\sum_{m=1}^{m_p} a_{-m}^{(p)} (\tau - \tau_p)^{-m}$. With the polylogarithm $\text{Li}_s(x) := \sum_{n \geq 1} x^n n^{-s}$, the polylog projection of f at τ_p is*

$$\hat{P}_{\tau_p} f(\tau) := \sum_{m=1}^{m_p} r_{f, \tau_p}^*(m) \text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)}), \quad (49)$$

its weights $r_{f, \tau_p}^(m)$ fixed by matching principal parts with f at τ_p ; and $\hat{P} := \sum_{\text{Im } \tau_p \geq 1} \hat{P}_{\tau_p}$.*

Lemma 4.6. *The weights $r_{f, \tau_p}^*(m)$, $1 \leq m \leq m_p$, exist and are unique, and $(I - \hat{P}_{\tau_p})f$ is holomorphic at τ_p .*

Proof. $\text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)})$ has a pole of order exactly m at τ_p , so matching the principal part of f is a triangular—hence uniquely solvable—linear system relating $r_{f, \tau_p}^*(1), \dots, r_{f, \tau_p}^*(m)$ to $a_{-1}^{(p)}, \dots, a_{-m}^{(p)}$; equal principal parts leave $(I - \hat{P}_{\tau_p})f$ regular at τ_p . $\square$

Definition 4.7. *With $\hat{P}$ as in Definition 4.5, let $c_{Rf}(n)$ be the Fourier coefficients of the pole-projected form $(I - \hat{P})f$ in its expansion valid for large $\text{Im } \tau$:*

$$(I - \hat{P})f(\tau) = \sum_n c_{Rf}(n) q^n, \quad q = e^{2\pi i \tau}. \quad (50)$$

Definition 4.8. $F(\infty)$ *is the space of meromorphic functions $g : \mathbb{H} \rightarrow \mathbb{C}$ with $g(\tau + 1) = g(\tau)$, finitely many poles in each compact subset of the strip $|\text{Re } \tau| \leq \frac{1}{2}$, and bounded pole order at $q = 0$.*

Lemma 4.9. $F_k \subset F(\infty)$ *and $(I - \hat{P})f \in F(\infty)$. Every pole of $(I - \hat{P})f$ has $\text{Im } \tau < 1$, with a maximal height $y_b < 1$ (set $y_b = 0$ if $(I - \hat{P})f$ is holomorphic).*

Proof. A weakly holomorphic form is T -periodic, meromorphic, and $O(e^{C\text{Im}\tau})$ at the cusp, so $F_k \subset F(\infty)$; the projected polylogs $\text{Li}_{1-m}(e^{2\pi i(\tau-\tau_p)})$ share these properties, so $(I - \hat{P})f \in F(\infty)$ (and leaves F_k , being non-modular). By Lemma 4.6, $\hat{P}$ cancels the principal part of every pole with $\text{Im}\tau \geq 1$, so the poles of $(I - \hat{P})f$ lie strictly below $\text{Im}\tau = 1$; finite in each compact set and accumulating only at $\text{Im}\tau = 0$, they have a maximal height $y_b < 1$. $\square$

The Fourier coefficients of a pole-projected form are controlled by its highest pole, an estimate we import from [3]:

Lemma 4.10. *With y_b as in Lemma 4.9 and m_b the maximal order among poles at height y_b , there is a $C > 0$ such that, for every $\varepsilon > 0$,*

$$C n^{m_b-1} e^{2\pi n y_b} \leq |c_{Rf}(n)| \leq e^{2\pi n(y_b+\varepsilon)}, \quad (51)$$

the upper bound for all large n , the lower for infinitely many n . In particular $|c_{Rf}(n)| e^{-2\pi n} \rightarrow 0$ geometrically, at base $e^{-2\pi(1-y_b-\varepsilon)}$.

Proof. This is the pole-height/order Fourier-growth estimate of [3] for a T -periodic meromorphic function, applied to $(I - \hat{P})f$ (Lemma 4.9). $\square$

For $n \neq 0$, define

$$G_{s,k}(n) := \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}. \quad (52)$$

Each pole with $\text{Im}\tau_p \geq 1$ contributes one projected polylog to $L^*(f, s)$, whose L-integral we now record.

Definition 4.11. *For a pole projected at z with $\text{Im}z > 1$, let $I_N(s, z)$ be the L-integral of the projected polylog $\text{Li}_{-N}(e^{2\pi i(\cdot-z)})$ in the minimal class $[\hat{\gamma}_f^S], [\hat{\gamma}_f^T]$ of Definition 4.3:*

$$I_N(s, z) := L^*(\text{Li}_{-N}(e^{2\pi i(\cdot-z)}), s; [\hat{\gamma}_f^S], [\hat{\gamma}_f^T]). \quad (53)$$

Lemma 4.12. *For $\text{Im}z > 1$, polylog inversion $\text{Li}_{-N}(w) = (-1)^{N+1} \text{Li}_{-N}(1/w)$ resums (53) into*

$$I_N(s, z) = \delta_{N,0} \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} \sum_{n \geq 1} n^N e^{2\pi i n z} G_{s,k}(-n), \quad (54)$$

convergent exponentially with base $e^{-2\pi(\text{Im}z-1)}$.

Proof. By Proposition 4.1 we may evaluate $I_N(s, z)$ in the $\delta \rightarrow 0$ representative of the minimal class; there the projected polylog — regular at i since $\text{Im}z > 1$ — has vanishing γ^S -integral and $\hat{\gamma}_f^T \rightarrow [i-1, i]$. On $[i-1, i]$ one has $|e^{2\pi i(\tau-z)}| > 1$, so the defining series of Li_{-N} diverges; inversion rewrites it as $(-1)^{N+1} \sum_{n \geq 1} n^N e^{2\pi i n(z-\tau)}$, convergent since $|e^{2\pi i(z-\tau)}| < 1$, and term-by-term integration against $\tilde{k}_T$ gives $\sum_n n^N e^{2\pi i n z} G_{s,k}(-n)$, the mode- $(-n)$ block of Theorem 3.2; for $N = 0$ the leftover -1 in $\text{Li}_0(w) = -1 - \text{Li}_0(1/w)$ contributes $\frac{1}{s} + \frac{i^k}{k-s}$. The rate is $e^{-2\pi(\text{Im}z-1)}$, from $\Gamma(s, -2\pi n) \sim (2\pi n)^{s-1} e^{2\pi n}$ against $|e^{2\pi i n z}| = e^{-2\pi n \text{Im}z}$. $\square$

Extracting similar results for L^* evaluated on the polar terms for poles on the line $\text{Im}(\tau) = 1$ requires more care. To do this away from the special point $z = i$, we find it helpful to define the following three functions:

$$\Pi_N(s, z) := \sum_{j=0}^{N+1} \frac{(-1)^{j+1} [(s-1)_{(j)} + i^k (k-s-1)_{(j)}]}{(2\pi)^{j+1}} \text{Li}_{j+1-N}(e^{2\pi i(z-i)}), \quad (55)$$

$$\Lambda_N(s, z) := \frac{\Gamma(s)}{(-2\pi)^s} \text{Li}_{s-N}(e^{2\pi iz}) + i^k \frac{\Gamma(k-s)}{(-2\pi)^{k-s}} \text{Li}_{k-s-N}(e^{2\pi iz}), \quad (56)$$

$$\begin{aligned} \mathcal{E}_N(s, z) := & - \sum_{n \geq 1} n^N e^{2\pi inz} \left[(s-1)_{(N+2)} \frac{\Gamma(s-N-2) - \Gamma(s-N-2, -2\pi n)}{(-2\pi n)^s} \right. \\ & \left. + i^k (k-s-1)_{(N+2)} \frac{\Gamma(k-s-N-2) - \Gamma(k-s-N-2, -2\pi n)}{(-2\pi n)^{k-s}} \right], \end{aligned} \quad (57)$$

where $(a)_{(j)} := a(a-1)\cdots(a-j+1)$ is the falling factorial ($(a)_{(0)} := 1$), Li_s the polylogarithm of Definition 4.5, and $\Gamma, \Gamma(a, x)$ the gamma and upper incomplete gamma, on principal branches [10].

Lemma 4.13. *With $\Pi_N, \Lambda_N, \mathcal{E}_N$ as in Eqs. (55)–(57), for $\text{Im } z > 1$ and every $s \in \mathbb{C}$,*

$$I_N(s, z) = \delta_{N,0} \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} [\Pi_N(s, z) + \Lambda_N(s, z) + \mathcal{E}_N(s, z)], \quad (58)$$

and the right side extends $I_N(s, \cdot)$ continuously to the edge $\text{Im } z = 1$ for $z \notin i + \mathbb{Z}$.

Proof. Throughout write $w := e^{2\pi i(z-i)}$ and $Q := e^{2\pi iz}$. Since $e^{2\pi i(z-i)} = e^{2\pi} e^{2\pi iz}$, these satisfy $w = e^{2\pi} Q$, with $|Q| = e^{-2\pi \text{Im } z}$ and $|w| = e^{-2\pi(\text{Im } z - 1)}$. For $\text{Im } z > 1$ both moduli are < 1 , so every series below converges absolutely and the rearrangements are unrestricted; Step 4 pushes this to the edge. The one bookkeeping fact used repeatedly: a summand carrying an extra $e^{2\pi n}$ combines with $e^{2\pi inz}$ into $w^n = e^{2\pi in(z-i)}$, while a summand without it keeps $e^{2\pi inz} = Q^n$.

By Lemma 4.12, when $\text{Im}(z) > 1$ we can safely conclude:

$$I_N(s, z) = \delta_{N,0} \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} \sum_{n \geq 1} n^N e^{2\pi inz} G_{s,k}(-n). \quad (59)$$

Step 1: peel $N+2$ orders off the incomplete gamma. Iterating $\Gamma(a, x) = x^{a-1} e^{-x} + (a-1)\Gamma(a-1, x)$ ([10, §8.8]) J times gives, for any integer $J \geq 1$,

$$\Gamma(s, x) = e^{-x} \sum_{j=0}^{J-1} (s-1)_{(j)} x^{s-1-j} + (s-1)_{(J)} \Gamma(s-J, x). \quad (60)$$

Take $J = N+2$, $x = -2\pi n$, and divide by $(-2\pi n)^s$. As $-1-j$ is an integer, $(-2\pi n)^{s-1-j}/(-2\pi n)^s = (-2\pi n)^{-1-j} = (-1)^{j+1} (2\pi n)^{-(j+1)}$, so

$$\frac{\Gamma(s, -2\pi n)}{(-2\pi n)^s} = \underbrace{e^{2\pi n} \sum_{j=0}^{N+1} \frac{(-1)^{j+1} (s-1)_{(j)}}{(2\pi n)^{j+1}}}_{\text{finite part}} + \underbrace{(s-1)_{(N+2)} \frac{\Gamma(s-N-2, -2\pi n)}{(-2\pi n)^s}}_{\text{tail}}. \quad (61)$$

Step 2: the finite part resums to Π_N . Multiply the finite part by $n^N e^{2\pi inz}$ and sum over n . The factor $e^{2\pi n} e^{2\pi inz} = w^n$, and $\sum_{n \geq 1} n^{-a} w^n = \text{Li}_a(w)$ at $a = j+1-N$ gives $\sum_{n \geq 1} n^{N-j-1} w^n = \text{Li}_{j+1-N}(w)$, whence

$$\sum_{n \geq 1} n^N e^{2\pi inz} e^{2\pi n} \sum_{j=0}^{N+1} \frac{(-1)^{j+1} (s-1)_{(j)}}{(2\pi n)^{j+1}} = \sum_{j=0}^{N+1} \frac{(-1)^{j+1} (s-1)_{(j)}}{(2\pi)^{j+1}} \text{Li}_{j+1-N}(w). \quad (62)$$

The i^k -partner repeats this with $(s-1)_{(j)} \mapsto i^k(k-s-1)_{(j)}$; the two together are $\Pi_N(s, z)$ of (55).

Step 3: the tail splits into Λ_N and $\mathcal{E}_N$. Write the reduced gamma as its $x \rightarrow -\infty$ value minus the remainder,

$$\Gamma(s-N-2, -2\pi n) = \Gamma(s-N-2) - [\Gamma(s-N-2) - \Gamma(s-N-2, -2\pi n)], \quad (63)$$

the bracket being the lower incomplete gamma $\gamma(s-N-2, -2\pi n)$. Carry each through the tail, summed against $n^N e^{2\pi i n z} = n^N Q^n$.

(a) *Value* $\rightarrow \Lambda_N$. The falling factorial telescopes the gamma, $(s-1)_{(N+2)}\Gamma(s-N-2) = (s-1)(s-2)\cdots(s-N-2)\Gamma(s-N-2) = \Gamma(s)$; with $n^N/(-2\pi n)^s = (-2\pi)^{-s}n^{N-s}$ and $\sum_{n \geq 1} n^{N-s}Q^n = \text{Li}_{s-N}(Q)$,

$$(s-1)_{(N+2)}\Gamma(s-N-2) \sum_{n \geq 1} \frac{n^N Q^n}{(-2\pi n)^s} = \frac{\Gamma(s)}{(-2\pi)^s} \text{Li}_{s-N}(Q), \quad (64)$$

the first term of Λ_N ((56)).

(b) *Remainder* $\rightarrow \mathcal{E}_N$. The bracket carries through verbatim to

$$-(s-1)_{(N+2)} \sum_{n \geq 1} n^N Q^n \frac{\Gamma(s-N-2) - \Gamma(s-N-2, -2\pi n)}{(-2\pi n)^s}, \quad (65)$$

the first term of $\mathcal{E}_N$ ((57)). The i^k -partners of (a),(b) give the second terms of $\Lambda_N, \mathcal{E}_N$. Collecting Steps 2–3 under the prefactor $(-1)^{N+1}$ yields (58).

Step 4: convergence and the edge. Only $\mathcal{E}_N$ needs an estimate. As $n \rightarrow \infty$, $x = -2\pi n \rightarrow -\infty$ and $\Gamma(\sigma, x) \sim x^{\sigma-1}e^{-x}$; with $\sigma = s-N-2$ this dominates the constant $\Gamma(s-N-2)$, so the numerator in (b) has size $(2\pi n)^{\text{Re } s-N-3}e^{2\pi n}$. Dividing by $|(-2\pi n)^s| = (2\pi n)^{\text{Re } s}$ and multiplying by n^N cancels the $\text{Re } s$, so the coefficient of Q^n is $O(n^{-3}e^{2\pi n})$. Since $e^{2\pi n}|Q|^n = e^{-2\pi n(\text{Im } z-1)} = |w|^n$, the terms of $\mathcal{E}_N$ are

$$O(n^{-3}|w|^n) \quad \text{uniformly on } \text{Im } z \geq 1. \quad (66)$$

As $|w| \leq 1$ there, $\mathcal{E}_N$ converges absolutely — on the edge $|w| = 1$ as well, by $\sum n^{-3}$. For continuity up to $\text{Im } z = 1$: Λ_N is a fixed combination of $\text{Li}_{s-N}(Q), \text{Li}_{k-s-N}(Q)$ with $|Q| \leq e^{-2\pi}$, hence continuous; $\mathcal{E}_N$ is continuous by the uniform bound; and in Π_N each $\text{Li}_{j+1-N}(w)$ is continuous on $|w| \leq 1$ away from its branch point $w = 1$. Finally $w = 1 \iff z - i \in \mathbb{Z}$, i.e. $z \in i + \mathbb{Z}$ — the excluded points. Hence the right side of (58) extends $I_N(s, \cdot)$ continuously to $\text{Im } z = 1$ for $z \notin i + \mathbb{Z}$. $\square$

The one edge point the formula misses is $z = i$, where Π_N is singular ($w = 1$); we compute $I_N(s, i)$ directly. Its finiteness rests on a relation among the Laurent coefficients of f at the S -fixed point.

Lemma 4.14. *Let $f \in F_k$ have a pole of order P at i , with principal part $\sum_{m=1}^P a_{-m}(\tau-i)^{-m}$. Modularity $f(-1/\tau) = \tau^k f(\tau)$ is equivalent to*

$$(-1)^r \sum_{m=r}^P a_{-m} \binom{m}{r} i^{m-r} = i^k \sum_{m=r}^P a_{-m} \binom{k}{m-r} (-i)^{m-r}, \quad r = 1, \dots, P. \quad (67)$$

The leading relation ($r = P$) is $((-1)^P - i^k)a_{-P} = 0$, so the pole order matches $k \bmod 4$ (P odd $\Leftrightarrow k \equiv 2$, P even $\Leftrightarrow 4 \mid k$); at $P = 1$ this is $(1+i^k)a_{-1} = 0$, and for a double pole the $r = 1$ relation is $a_{-1} = \frac{i(k-2)}{2}a_{-2}$.

Proof. Write $\tau = i + \varepsilon$. Then $-1/\tau - i = -\varepsilon/(1 - i\varepsilon)$, so the principal part of $f(-1/\tau)$ is $\sum_m a_{-m}(i - \varepsilon^{-1})^m$ and that of $\tau^k f(\tau)$ is $i^k(1 - i\varepsilon)^k \sum_m a_{-m}\varepsilon^{-m}$. Matching the coefficient of ε^{-r} through $(i - \varepsilon^{-1})^m = \sum_j \binom{m}{j} i^{m-j}(-1)^j \varepsilon^{-j}$ and $(1 - i\varepsilon)^k = \sum_l \binom{k}{l} (-i)^l \varepsilon^l$ gives (63). $\square$

Definition 4.15. For $N \geq 0$ and any cutoff $J \geq N + 2$, with $H_N := \sum_{r=1}^N \frac{1}{r}$ and $c_j^+(s) := (s-1)_{(j)} + i^k(k-s-1)_{(j)}$, set

$$\mathcal{E}_N^+(s) := \sum_{n \geq 1} n^N e^{2\pi n} \left[(s-1)_{(J)} \frac{\Gamma(s-J, 2\pi n)}{(2\pi n)^s} + i^k(k-s-1)_{(J)} \frac{\Gamma(k-s-J, 2\pi n)}{(2\pi n)^{k-s}} \right], \quad (68)$$

and then

$$\text{FP}_N(s, k) := \frac{c_N^+(s) H_N}{(2\pi)^{N+1}} + \sum_{\substack{0 \leq j \leq J \\ j \neq N}} \frac{c_j^+(s) \zeta(j+1-N)}{(2\pi)^{j+1}} + \frac{(-1)^N N!}{(2\pi)^{N+1}} \sum_{t=0}^{N-1} \frac{(-1)^t (s-1)_{(t)}}{t! (N-t)} + \mathcal{E}_N^+(s). \quad (69)$$

Lemma 4.16. Let $f \in F_k$ have a pole of order P at i with principal part $\sum_{m=1}^P a_{-m}(\tau-i)^{-m}$, and $\hat{P}_i f$ its projection. The pole's contribution to $L^*(f, s)$ in the minimal class is

$$L^*(\hat{P}_i f, s) = -2\pi i a_{-1} \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{m=1}^P \frac{(-2\pi i)^m a_{-m}}{(m-1)!} \text{FP}_{m-1}(s, k), \quad (70)$$

entire in s but for the simple poles at $s = 0, k$ carried by a_{-1} ; the individually divergent contour pieces cancel by the relations of Lemma 4.14.

Proof. Evaluate on the minimal-class contour with basepoint $i(1 + \delta)$ and let $\delta \rightarrow 0$. On the T -segment the pole at i lies below the contour; inverting each $\text{Li}_{1-m}(e^{2\pi i(\tau-i)})$ and integrating against $\tilde{k}_T$ yields the harmonic-, ζ -, and $\mathcal{E}^+$ -terms of FP_{m-1} together with a part $\sim \delta^{-(m-1)}, \dots, \log \delta$; the S -segment, passing to the left of i , gives the mirror divergences. Summed over m with weights $r^*(m) = (-2\pi i)^m a_{-m}/(m-1)!$, the divergent parts form a linear combination of the a_{-m} that vanishes precisely by the modularity relations of Lemma 4.14, leaving the FP -sum of (66); the simple part ($m = 1$) additionally carries the -1 -leftover pole $-2\pi i a_{-1}(\frac{1}{s} + \frac{i^k}{k-s})$, as in Lemma 4.12. The cutoff J is immaterial: raising it moves terms between the ζ -sum and $\mathcal{E}_N^+$, which remains convergent as in Lemma 4.13. $\square$

Theorem 4.17. Let $f \in F_k$ have poles τ_p of order P_p with $\text{Im } \tau_p \geq 1$ in the strip $|\text{Re } \tau| \leq \frac{1}{2}$, pole-projected Fourier coefficients $c_{Rf}(n)$ (Definition 4.7), and projection weights $r_{f, \tau_p}^*(m)$ (Definition 4.5). In the minimal class,

$$L^*(f, s) = -c_{Rf}(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_{Rf}(n) G_{s,k}(n) + \sum_{\tau_p \neq i} \sum_{m=1}^{P_p} r_{f, \tau_p}^*(m) I_{m-1}(s, \tau_p) + L^*(\hat{P}_i f, s), \quad (71)$$

with I_{m-1} from Lemmas 4.12–4.13 and the last term (present only for a pole at i) from Lemma 4.16. The right side is entire in s apart from at most simple poles at $s = 0$ and $s = k$, and reduces to the weakly-holomorphic L^* of Theorem 3.2 when f is holomorphic on $\mathbb{H}$.

Proof. Split $f = (I - \hat{P})f + \hat{P}f$. By Lemma 4.10 the Fourier series of the pole-projected part converges on $\hat{\gamma}_f^T$; Theorem 3.2 applied term by term gives its L^* as the first two sums. The projection $\hat{P}f = \sum_{\tau_p} \sum_m r_{f, \tau_p}^*(m) \text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)})$ contributes, for each $\tau_p \neq i$, the L-integrals $I_{m-1}(s, \tau_p)$ of Definition 4.11, evaluated by Lemmas 4.12–4.13; the pole at i contributes $L^*(\hat{P}_i f, s)$ of Lemma 4.16. Entirety and the pole locations follow from those of $G_{s,k}$ and the $\delta_{N,0}$ terms; when f is holomorphic on $\mathbb{H}$, $\hat{P}f = 0$ and $c_{Rf} = c_f$, recovering Theorem 3.2. $\square$

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